

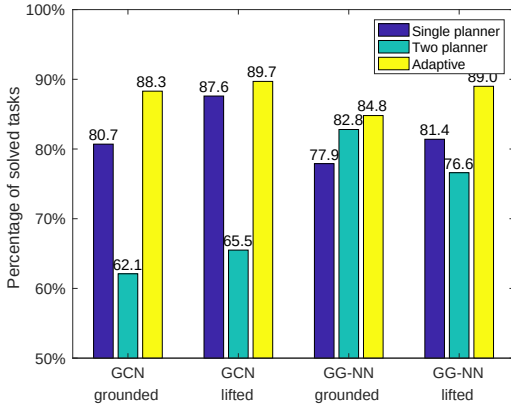


Figure 4: Percentage of solved tasks in the test set. Delfi split; two planners.

One similarly sees that adaptive scheduling consistently improves over single-planner scheduling.

Multiple planners; Delfi split; simple greedy: The success of adaptive scheduling raises much interest in the use of more than one planner. A straightforward extension of the adaptive model g is to select more than two planners; but as the number of planners increases, the model becomes more and more complex and challenging to train. Instead, we explore offline, non-machine learning selection methods.

In Table 4 we consider the following approach: k planners are selected (independent of the task) and each is allotted T/k time. The first selected planner is the best performer on the training+validation set in the allotted time. The next planners are chosen in the same manner on the portion of the set not solved by previously selected planners. The top four rows of Table 4 report the results. For reference, the next four rows report the ceiling performance were the planners selected from the test set.

If $k = 2$ planners are used, the coverage is 85.5%, lower than that achievable by the proposed adaptive scheduling approach, 89.7% (see Figure 4). If one is willing to use more planners, a better result is achieved with $k = 3$. However, the coverage does not increase monotonically with k . Moreover, the performance appears to be much sensitive to the change of k . Hence, it is challenging to find the best k .

Table 4: Percentage of solved tasks in the test set. Offline method. Delfi split. Compare with performance in Figure 4.

Method	Solved
(Greedy) Best 2 planners from train set	85.5%
(Greedy) Best 3 planners from train set	92.4%
(Greedy) Best 4 planners from train set	89.7%
(Greedy) Best 5 planners from train set	87.6%
(Greedy oracle) Best 2 planners from test set	93.8%
(Greedy oracle) Best 3 planners from test set	93.8%
(Greedy oracle) Best 4 planners from test set	93.1%
(Greedy oracle) Best 5 planners from test set	92.4%
Fast Downward Stone Soup	92.4%

Multiple planners; multiple splits; FDSS: A more sophisticated offline planner selection approach is Fast Downward Stone Soup (?), wherein the selection of planners and their allotted time are optimized according to a certain score function. Its result is reported in the last row of Table 4. This result is achieved with four planners. The coverage coincides with the above greedy approach when taking $k = 3$.

Conclusion

Graphs encode the structural information of a planning task. In this work, we have proposed a graph neural network approach for online planner selection. This approach outperforms Delfi, the winner of the Optimal Track of IPC 2018, which treats a planning task as an image and applies CNNs for selecting candidate planners. Our appealing results are owing to the representation power of GNNs that address the lack of permutation invariance and the negligence of node-labeling information in CNNs.

We have also proposed an adaptive scheduling approach to compensate the inaccuracy of a single predictive model, through offering a chance for switching planners at halftime, conditioned on the performance of the previously selected one. Such an adaptive approach consistently increases the number of solved tasks.

Overall, it appears that the lifted graph version is advantageous over the grounded one, because of consistently better performance. However, on average they are larger in size and some are particularly enormous. Moreover, the size distribution is highly skewed in both versions. These factors impose substantial challenges for the batch training of the neural networks. An avenue of future research is to investigate more efficient and scalable training approaches.

We have seen that the use of multiple planners beyond two may improve the performance. Another line of future work is to extend the proposed adaptive scheduling to more than two planners. In this case, the training set construction becomes more and more complex and the model may be challenging to train. Additionally, a principled approach of setting the right number of planners is to be developed.

Acknowledgments

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